

Initiating Search

November 5, 2025, 8:20 PM

• Search:

CAS Lexicon Search:

Fischer esterification - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (263)	 References	View Results
Exported: Retrieved Related Reaction Results (721)	 Reactions	View Results

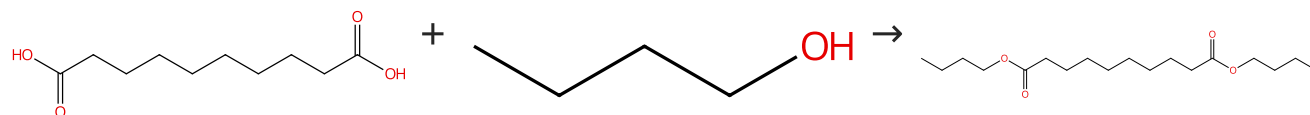
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 Reactions (50)[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (93) Suppliers (201) Suppliers (95)

31-614-CAS-45919371

Steps: 1 Yield: 100%

Dibutyl sebacate as PVC ecoplasticizer- economic synthesis and functional properties

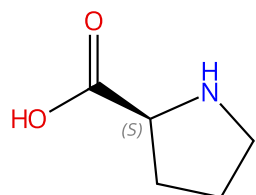
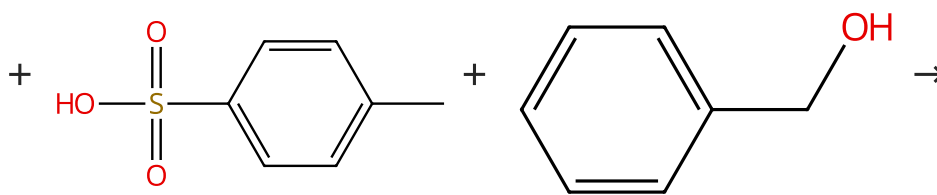
By: Tracz, Anna; et al

Royal Society Open Science (2025), 12(3), 241984.

1.1 Catalysts: Triethylammonium hydrogen sulfate; 2 h, 110 °C

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (-) Suppliers (168) Suppliers (63) Suppliers (149)Multi-component
structure image
available in CAS
SciFinder Suppliers (3)

31-356-CAS-21995822

Steps: 1 Yield: 100%

Preparation of enantiopure methionine, arginine, tryptophan, and proline benzyl esters in green ethers by Fischer-Speier reaction

By: Bolchi, Cristiano; et al

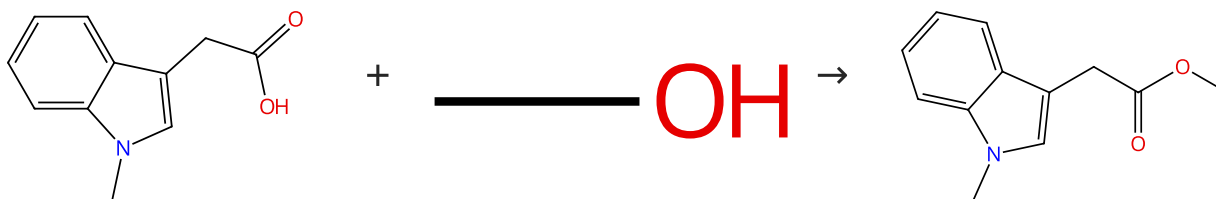
Amino Acids (2018), 50(9), 1261-1268.

1.1 Solvents: Cyclohexane; 4 h, reflux

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (65)

Suppliers (444)

Suppliers (35)

31-356-CAS-23549370

Steps: 1 Yield: 100%

Chemo- and Site-Selective Fischer Esterification Catalyzed by B(C₆F₅)₃

By: Li, Ya-Lan; et al

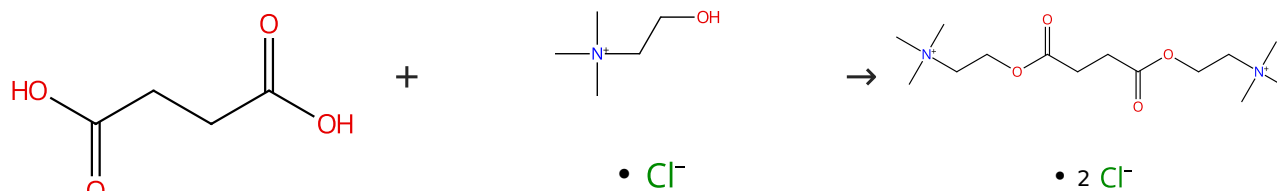
Asian Journal of Organic Chemistry (2021), 10(6), 1424-1427.

 1.1 Catalysts: Tris(pentafluorophenyl)borane
 Solvents: Methanol; 100 h, 24 °C

Experimental Protocols

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (150)

Suppliers (127)

Suppliers (33)

31-614-CAS-41123097

Steps: 1 Yield: 100%

Green and efficient deep eutectic solvent mediated one-step synthesis of Suxamethonium Chloride as Active Pharmaceutical Ingredient

By: Sagandira, Claudiu R.; et al

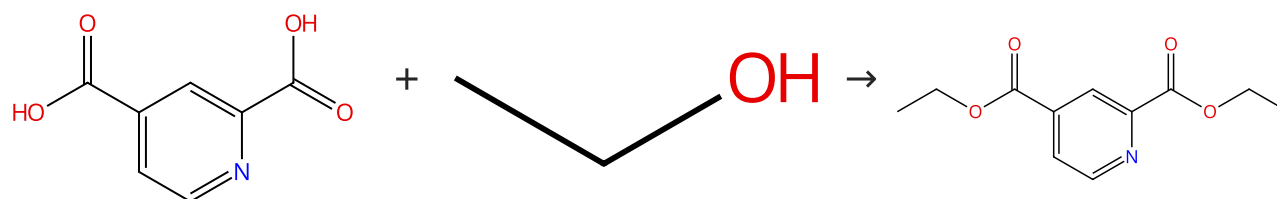
Sustainable Chemistry and Pharmacy (2024), 39, 101617.

1.1 Catalysts: Zinc chloride, Sulfuric acid; 4 h, 150 °C

Experimental Protocols

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (90)

Suppliers (618)

Suppliers (64)

31-356-CAS-13885913

Steps: 1 Yield: 100%

Modular synthesis and biological activity of pyridyl-based analogs of the potent Class I Histone Deacetylase Inhibitor Largazole

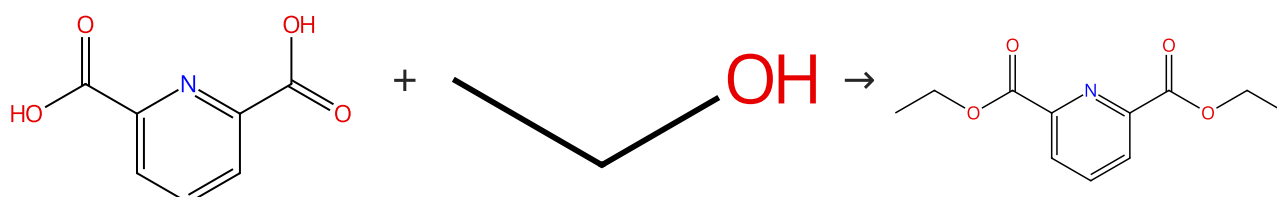
By: Clausen, Dane J.; et al

Bioorganic & Medicinal Chemistry (2015), 23(15), 5061-5074.

 1.1 Reagents: *p*-Toluenesulfonic acid
 Solvents: Ethanol; 80 °C

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (104)

Suppliers (618)

Suppliers (70)

31-356-CAS-5194663

Steps: 1 Yield: 100%

Modular synthesis and biological activity of pyridyl- based analogs of the potent Class I Histone Deacetylase Inhibitor Largazole

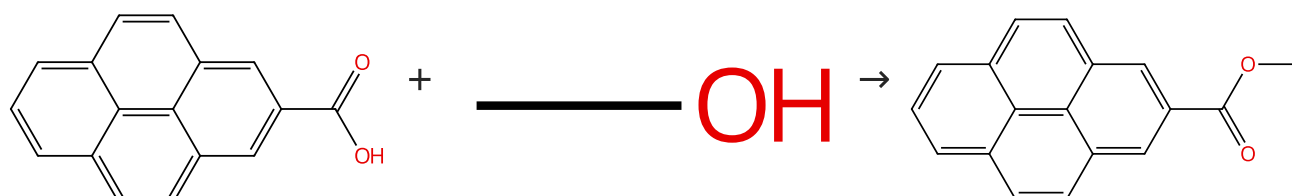
By: Clausen, Dane J.; et al

Bioorganic & Medicinal Chemistry (2015), 23(15), 5061-5074.

1.1 **Reagents:** *p*-Toluenesulfonic acid
Solvents: Ethanol; 80 °C

Scheme 7 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (44)

Suppliers (444)

Suppliers (5)

31-356-CAS-5784222

Steps: 1 Yield: 99%

Intramolecular excimer formation and delayed fluorescence in sterically constrained pyrene dimers

By: Benniston, Andrew C.; et al

Chemistry - A European Journal (2007), 13(16), 4665-4674.

1.1 **Reagents:** Sulfuric acid
Solvents: Methanol, Water; 24 h, reflux; reflux → rt

Experimental Protocols

Scheme 8 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (97)

Suppliers (444)

Suppliers (53)

31-356-CAS-17410233

Steps: 1 Yield: 99%

Efficient flow Fischer esterification of carboxylic acids with alcohols using sulfonic acid-functionalized silica as supported catalyst

By: Furuta, Akihiro; et al

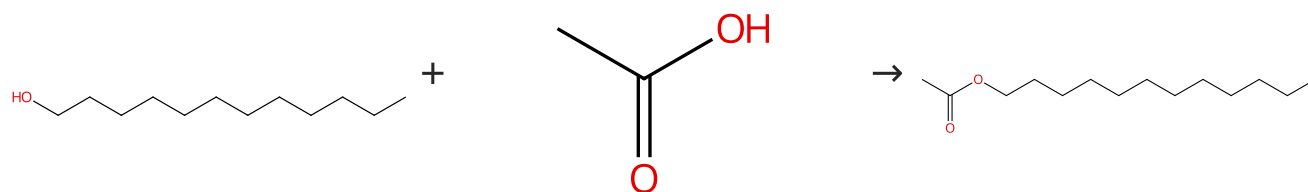
Bulletin of the Chemical Society of Japan (2017), 90(5), 607-612.

1.1 **Catalysts:** 2-Hydroxy-3-[3-(trihydroxysilyl)propoxy]-1-propanesulfonic acid (silica gel supported)
Solvents: Methanol, Tetrahydrofuran; 26 min, 75 psi, 110 °C

Experimental Protocols

Scheme 9 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (99)

Suppliers (324)

Suppliers (65)

31-356-CAS-11905528

Steps: 1 Yield: 99%

Application of triethylammonium salts as ionic liquid catalyst and medium for fischer esterification

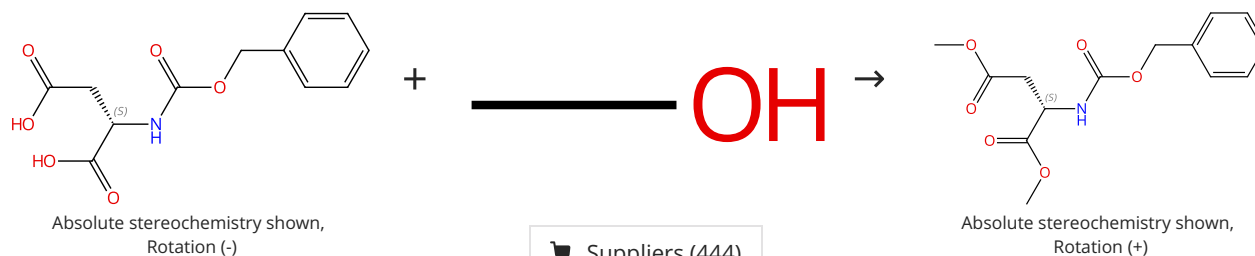
1.1 **Catalysts:** Triethylammonium sulfate
Solvents: Triethylammonium sulfate; 4 h, 90 °C

By: Ganeshpure, Pralhad A.; et al

ARKIVOC (Gainesville, FL, United States) (2007), (8), 273-278.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (78)

Suppliers (444)

Suppliers (37)

31-356-CAS-23549573

Steps: 1 Yield: 99%

Chemo- and Site-Selective Fischer Esterification Catalyzed by B(C₆F₅)₃

1.1 **Catalysts:** Tris(pentafluorophenyl)borane
Solvents: Methanol; 100 h, 24 °C

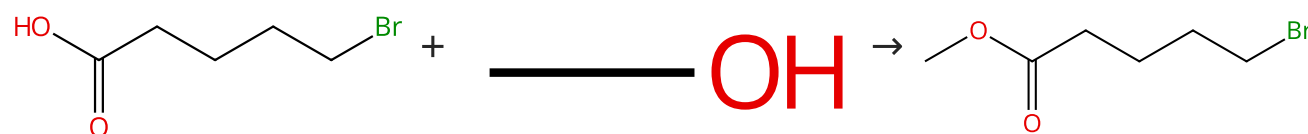
By: Li, Ya-Lan; et al

Experimental Protocols

Asian Journal of Organic Chemistry (2021), 10(6), 1424-1427.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (75)

Suppliers (444)

Suppliers (84)

31-356-CAS-17410228

Steps: 1 Yield: 99%

Efficient flow Fischer esterification of carboxylic acids with alcohols using sulfonic acid-functionalized silica as supported catalyst

1.1 **Catalysts:** 2-Hydroxy-3-[3-(trihydroxysilyl)propoxy]-1-propanesulfonic acid (silica gel supported)
Solvents: Methanol; 30 min, 75 psi, 110 °C

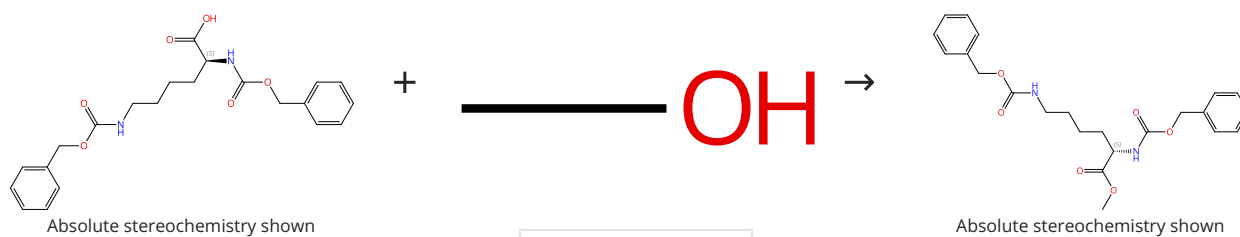
By: Furuta, Akihiro; et al

Experimental Protocols

Bulletin of the Chemical Society of Japan (2017), 90(5), 607-612.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (73)

Suppliers (444)

Suppliers (28)

31-356-CAS-23554265

Steps: 1 Yield: 99%

Chemo- and Site-Selective Fischer Esterification Catalyzed by $B(C_6F_5)_3$

By: Li, Ya-Lan; et al

Asian Journal of Organic Chemistry (2021), 10(6), 1424-1427.

1.1 **Catalysts:** Tris(pentafluorophenyl)borane
Solvents: Methanol; 100 h, 24 °C

Experimental Protocols

Scheme 13 (2 Reactions)

Steps: 1 Yield: 97-99%



Suppliers (117)

Suppliers (444)

Suppliers (103)

31-356-CAS-17410219

Steps: 1 Yield: 99%

Efficient flow Fischer esterification of carboxylic acids with alcohols using sulfonic acid-functionalized silica as supported catalyst

By: Furuta, Akihiro; et al

Bulletin of the Chemical Society of Japan (2017), 90(5), 607-612.

1.1 **Catalysts:** 2-Hydroxy-3-[3-(trihydroxysilyl)propoxy]-1-propanesulfonic acid (silica gel supported)
Solvents: Methanol; 30 min, 75 psi, 110 °C

Experimental Protocols

31-356-CAS-14723882

Steps: 1 Yield: 97%

Novel Bronsted acidic deep eutectic solvent as reaction media for esterification of carboxylic acid with alcohols

By: De Santi, Valerio; et al

Tetrahedron Letters (2012), 53(38), 5151-5155.

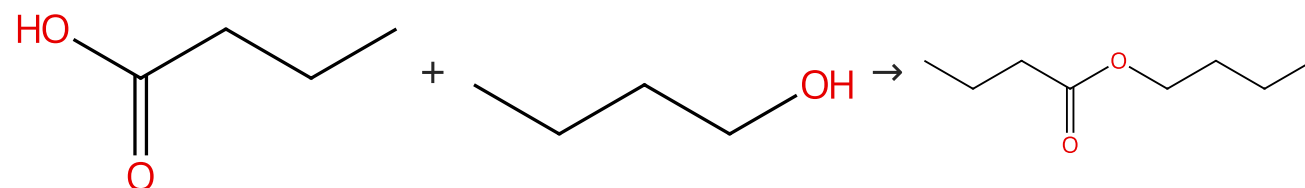
1.1 **Catalysts:** *p*-Toluenesulfonic acid, Cyclohexanaminium, *N,N,N*-trimethyl-, methanesulfonate (1:1); 10 min, rt → 60 °C

1.2 2 h, rt → 60 °C

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

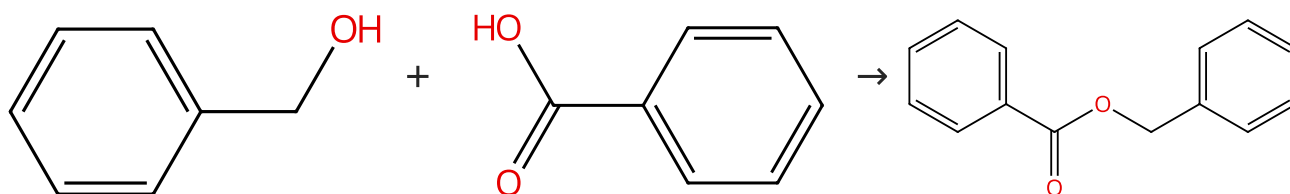
Suppliers (201)

Suppliers (61)

31-356-CAS-7922106	Steps: 1 Yield: 99%	Application of triethylammonium salts as ionic liquid catalyst and medium for fischer esterification
1.1 Catalysts: Triethylammonium sulfate Solvents: Triethylammonium sulfate; 4 h, 90 °C		By: Ganeshpure, Pralhad A.; et al ARKIVOC (Gainesville, FL, United States) (2007), (8), 273-278.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (149)

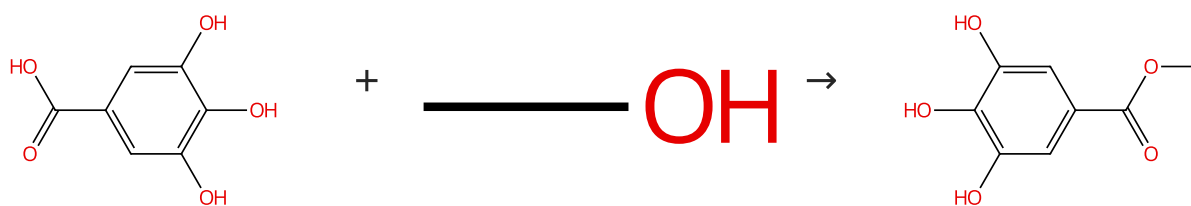
Suppliers (175)

Suppliers (119)

31-356-CAS-5370891	Steps: 1 Yield: 99%	Benzyl benzoate and dibenzyl ether from of benzoic acid and benzyl alcohol under microwave irradiation using a SiO ₂ -SO ₃ H catalyst
1.1 Catalysts: Silica (sulfonated); 5 min, 70 °C		By: Barbosa, Sandro L.; et al
Experimental Protocols		Catalysis Communications (2015), 68, 97-100.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (109)

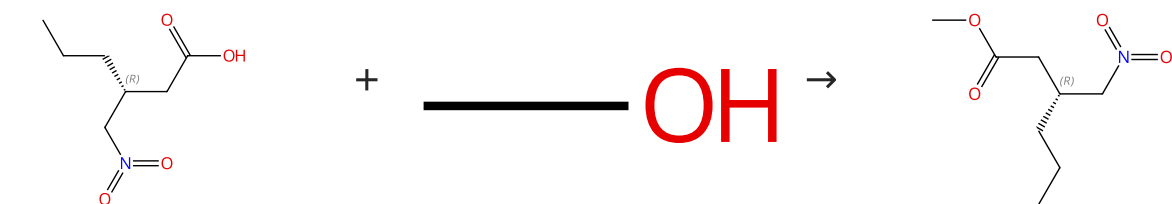
Suppliers (444)

Suppliers (98)

31-614-CAS-24697384	Steps: 1 Yield: 99%	Continuous-Flow Preparation of Benzotropolones: Combined Batch and Flow Synthesis of Epigenetic Modulators of the (JmJ C)-Containing Domain
1.1 Reagents: Sulfuric acid Solvents: Water; 34 min, 80 °C		By: Souto, Jose A.
Experimental Protocols		ChemistrySelect (2021), 6(39), 10717-10721.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 98%



Absolute stereochemistry shown

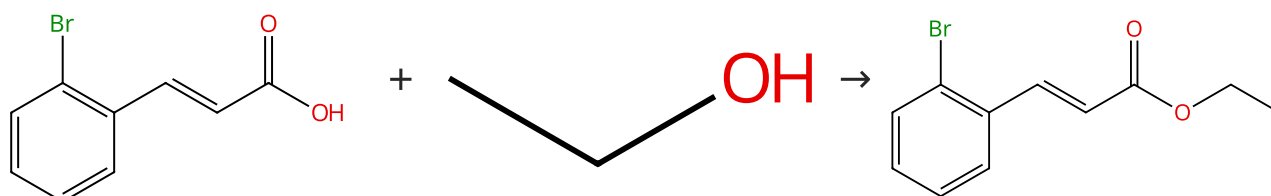
Absolute stereochemistry shown

Suppliers (444)

31-356-CAS-20975044	Steps: 1 Yield: 98%	Process for asymmetric synthesis of (R)-4-propyldihydrofuran-2(3H)-one Assignee: Procos S.p.A. World Intellectual Property Organization, WO2019239341 A1 2019-12-19 PatentPak available
1.1 Catalysts: Sulfuric acid Solvents: Methanol; 2 h, 40 °C		

Scheme 18 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (41)

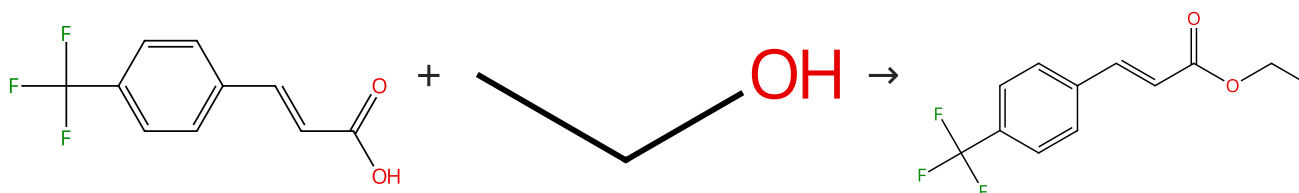
Suppliers (618)

Suppliers (39)

31-614-CAS-24575619	Steps: 1 Yield: 98%	Solvent role in the lipase-catalysed esterification of cinnamic acid and derivatives. Optimisation of the biotransformation conditions By: Suarez-Escobedo, Laura; et al Tetrahedron (2021), 81, 131873.
1.1 Reagents: Sulfuric acid Solvents: Ethanol, Water; 24 h, 82 °C		
Experimental Protocols		

Scheme 19 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (33)

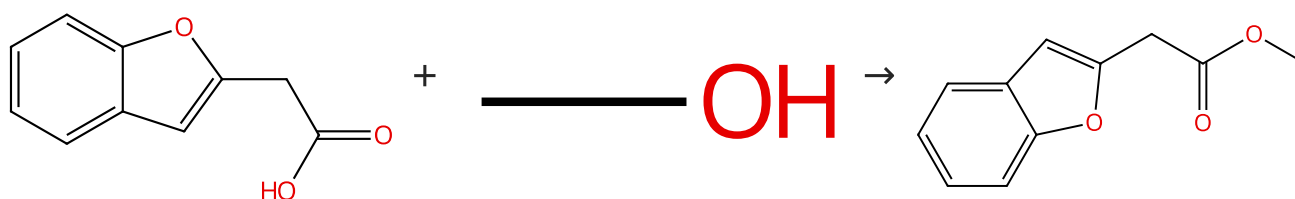
Suppliers (618)

Suppliers (36)

31-614-CAS-24575614	Steps: 1 Yield: 98%	Solvent role in the lipase-catalysed esterification of cinnamic acid and derivatives. Optimisation of the biotransformation conditions By: Suarez-Escobedo, Laura; et al Tetrahedron (2021), 81, 131873.
1.1 Reagents: Sulfuric acid Solvents: Ethanol, Water; 22 h, 82 °C		
Experimental Protocols		

Scheme 20 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (68)

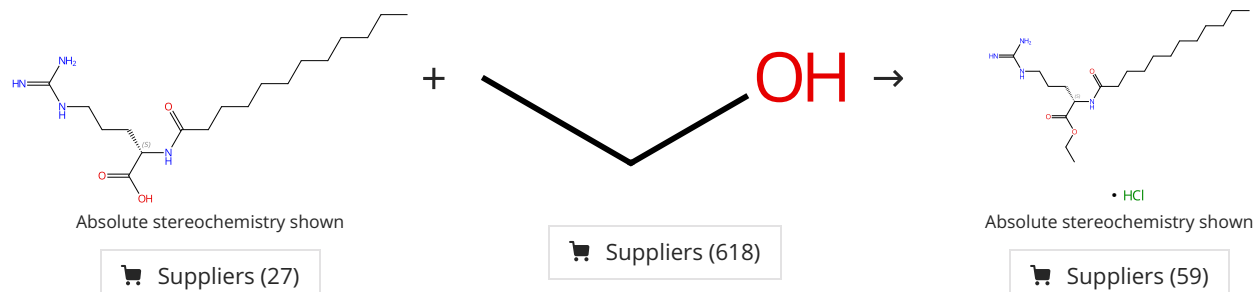
Suppliers (444)

Suppliers (32)

31-356-CAS-23548860	Steps: 1 Yield: 98%	Chemo- and Site-Selective Fischer Esterification Catalyzed by B(C ₆ F ₅) ₃
1.1 Catalysts: Tris(pentafluorophenyl)borane Solvents: Methanol; 100 h, 24 °C		By: Li, Ya-Lan; et al
Experimental Protocols		Asian Journal of Organic Chemistry (2021), 10(6), 1424-1427.

Scheme 21 (1 Reaction)

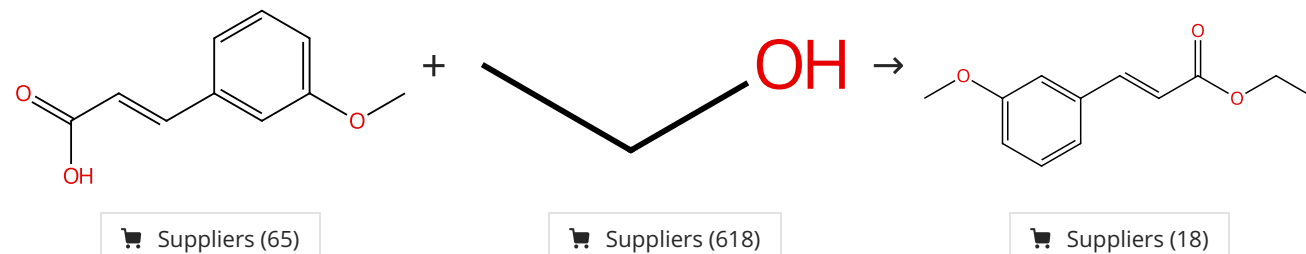
Steps: 1 Yield: 98%



31-614-CAS-46463477	Steps: 1 Yield: 98%	In situ synthesis of carboxylic ester compounds
1.1 Reagents: Sulfuric acid, Magnesium chloride Solvents: Ethanol; rt; 12 h, 30 °C; 30 °C → rt		Assignee: Woodward Bio Corp.
1.2 Reagents: Sodium hydroxide Solvents: Water; pH 4 - 6, rt		Korea, Republic of, KR2024055566 A 2024-04-29
1.3 Reagents: Hydrochloric acid Solvents: Water; pH 5 - 6, rt		PatentPak available

Scheme 22 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-24575601	Steps: 1 Yield: 98%	Solvent role in the lipase-catalysed esterification of cinnamic acid and derivatives. Optimisation of the biotransformation conditions
1.1 Reagents: Sulfuric acid Solvents: Ethanol, Water; 22 h, 82 °C		By: Suarez-Escobedo, Laura; et al
Experimental Protocols		Tetrahedron (2021), 81, 131873.

Scheme 23 (1 Reaction)

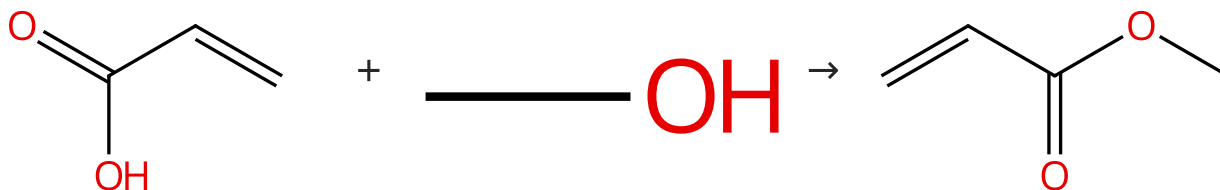
Steps: 1 Yield: 98%



31-614-CAS-42419509	Steps: 1 Yield: 98%	Solvent Attenuation of London Dispersion in Polycyclic Aromatic Stacking By: Elmi, Alex; et al Angewandte Chemie, International Edition (2024), 63(45), e202412056.
1.1 Catalysts: Sulfuric acid; 18 h, rt → reflux		

Scheme 24 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (65)

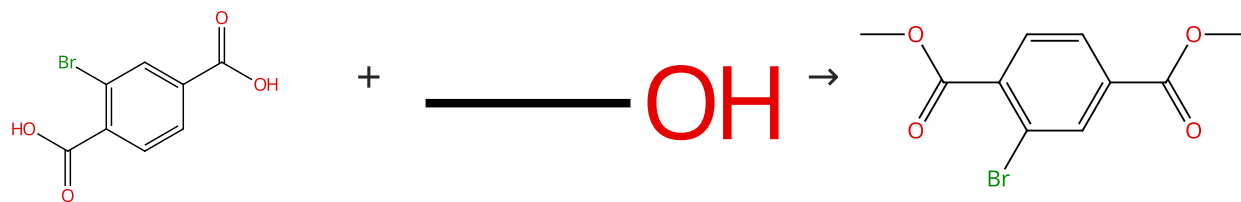
Suppliers (444)

Suppliers (60)

31-356-CAS-20966883	Steps: 1 Yield: 98%	Second-Generation meta-Phenolsulfonic Acid-Formaldehyde Resin as a Catalyst for Continuous-Flow Esterification By: Hu, Hao; et al Organic Letters (2020), 22(1), 160-163.
1.1 Catalysts: Benzenesulfonic acid, 4-hydroxy-, polymer with formaldehyde (second-generation m-phenolsulfonic acid-formaldehyde resin (PAFRII)); 90 °C		
Experimental Protocols		

Scheme 25 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (90)

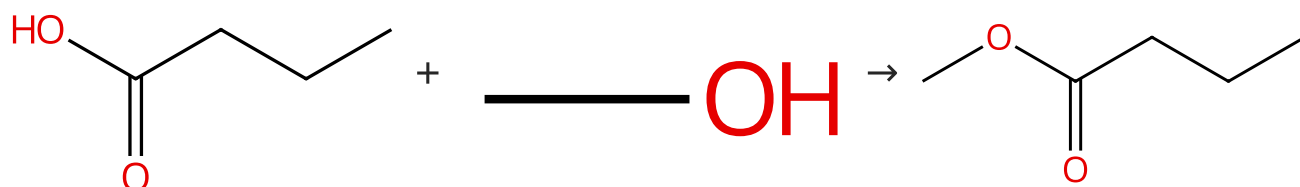
Suppliers (444)

Suppliers (70)

31-356-CAS-10035119	Steps: 1 Yield: 98%	Process for the preparation of vinylphthalates useful for preparing aminocarbonylstyrene and liquid crystalline polymers Assignee: Toray Chemical Korea, Inc. Korea, Republic of, KR2015033493 A 2015-04-01 PatentPak available
1.1 Reagents: Sulfuric acid; 5 h, reflux		

Scheme 26 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (82)

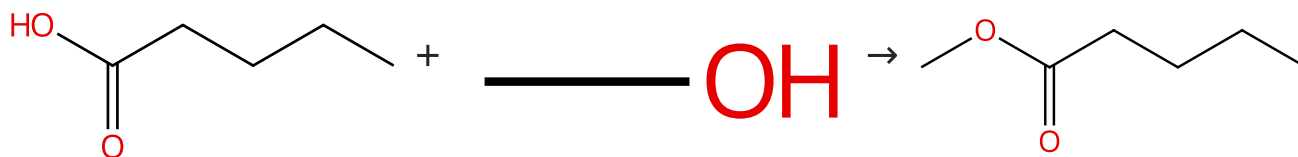
Suppliers (444)

Suppliers (79)

31-356-CAS-7714838	Steps: 1 Yield: 98%	Solvent-free synthesis of long chain aliphatic acid methyl esters in bronsted acidic ionic liquids at room temperature
1.1 Solvents: 1 <i>H</i> -Imidazolium, 1-methyl-3-[(4-sulfophenyl)methyl]-, sulfate (1:1); 3 h, rt		By: Li, Xinzhong; et al Jingxi Huagong (2009), 26(11), 1141-1144.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (88)

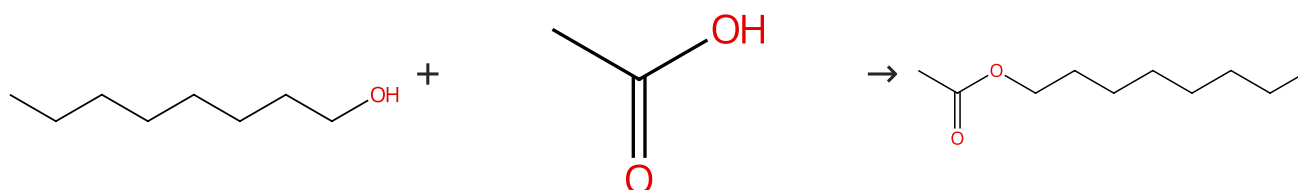
Suppliers (444)

Suppliers (59)

31-356-CAS-9851690	Steps: 1 Yield: 98%	Solvent-free synthesis of long chain aliphatic acid methyl esters in bronsted acidic ionic liquids at room temperature
1.1 Solvents: 1 <i>H</i> -Imidazolium, 1-methyl-3-[(4-sulfophenyl)methyl]-, sulfate (1:1); 3 h, rt		By: Li, Xinzhong; et al Jingxi Huagong (2009), 26(11), 1141-1144.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (132)

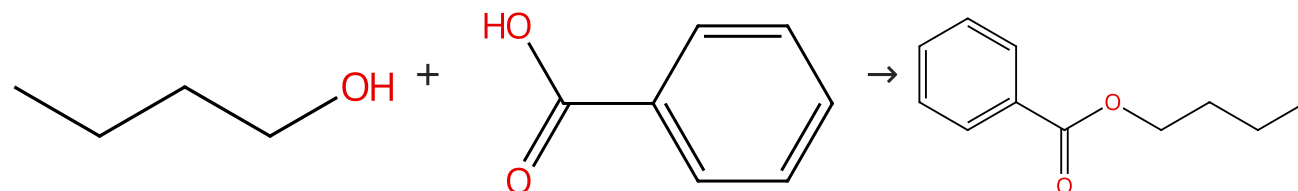
Suppliers (324)

Suppliers (83)

31-356-CAS-7647230	Steps: 1 Yield: 98%	Application of triethylammonium salts as ionic liquid catalyst and medium for fischer esterification
1.1 Catalysts: Triethylammonium sulfate Solvents: Triethylammonium sulfate; 4 h, 90 °C		By: Ganeshpure, Pralhad A.; et al ARKIVOC (Gainesville, FL, United States) (2007), (8), 273-278.

Scheme 29 (2 Reactions)

Steps: 1 Yield: 97-98%



Suppliers (201)

Suppliers (175)

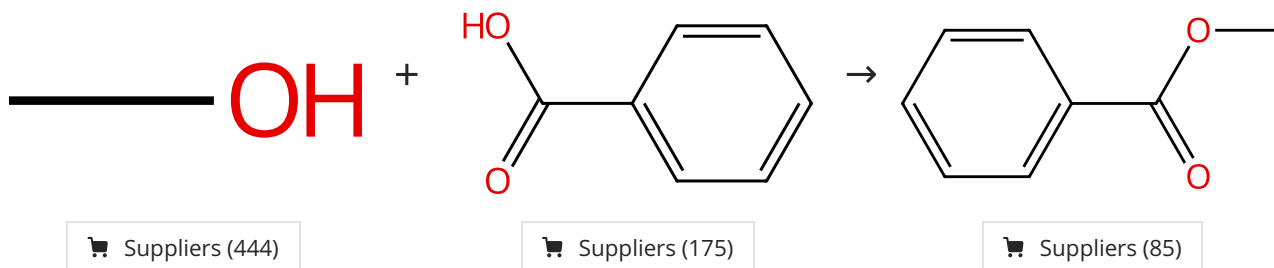
Suppliers (76)

31-614-CAS-35672318	Steps: 1 Yield: 98%	Fischer Esterification of Benzoic Acid and Polyhydric Alcohols Catalyzed by Basic Ionic Liquids of Bisimidazolium Tungstates
1.1 Catalysts: 2764700-49-2; 1 h, 0.5 M Pa, 150 °C		By: Wu, Shi; et al
Experimental Protocols		ChemistrySelect (2023), 8(6), e202204112.

31-356-CAS-14564293	Steps: 1 Yield: 97%	Novel Bronsted-acidic ionic liquids based on benzothiazolium cations as catalysts for esterification reactions
1.1 Catalysts: Benzothiazole 1,1,1-trifluoromethanesulfonate (1:1); 8 h, 110 °C; 110 °C → rt		By: Zhou, Xian Si; et al Journal of the Serbian Chemical Society (2011), 76(12), 1607-1615.

Scheme 30 (1 Reaction)

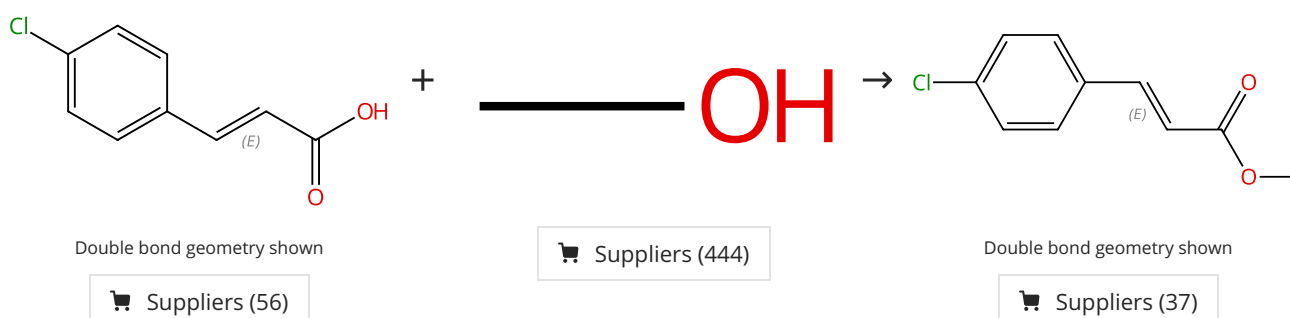
Steps: 1 Yield: 98%



31-356-CAS-12165408	Steps: 1 Yield: 98%	Novel Bronsted-acidic ionic liquids based on benzothiazolium cations as catalysts for esterification reactions
1.1 Catalysts: Benzothiazole methanesulfonate (1:1); 8 h, 120 °C; 120 °C → rt		By: Zhou, Xian Si; et al Journal of the Serbian Chemical Society (2011), 76(12), 1607-1615.

Scheme 31 (1 Reaction)

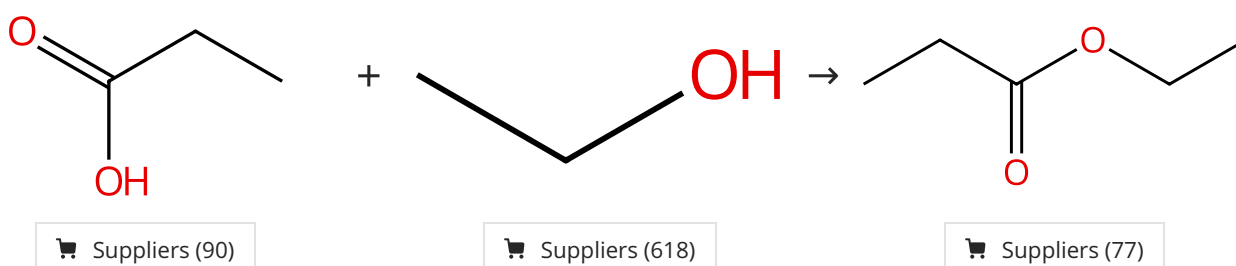
Steps: 1 Yield: 98%



31-356-CAS-20621274	Steps: 1 Yield: 98%	Antimicrobial activity of 4-chlorocinnamic acid derivatives
1.1 Catalysts: Sulfuric acid; 3 - 24 h, reflux		By: Silva, Rayanne H. N.; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water; neutralized, rt		BioMed Research International (2019), 3941242pp..
Experimental Protocols		

Scheme 32 (1 Reaction)

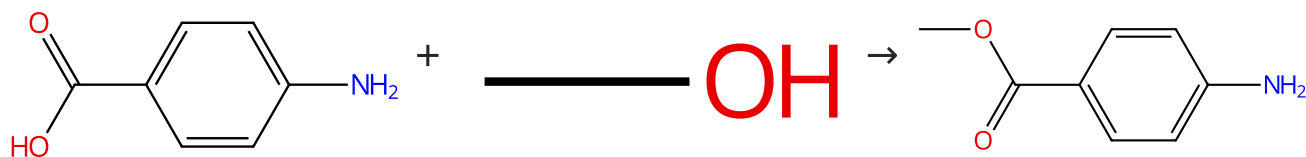
Steps: 1 Yield: 98%



31-356-CAS-4329571	Steps: 1 Yield: 98%	Study on the catalytic behaviors of expansible graphite in the synthesis of acetic ester and propionic ester
1.1 Catalysts: Graphite Solvents: Water; 0.75 h, 98 °C		By: Lin, Rui-Nian; et al Organic Chemistry: An Indian Journal (2013), 9(3), 92-96.

Scheme 33 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (129)

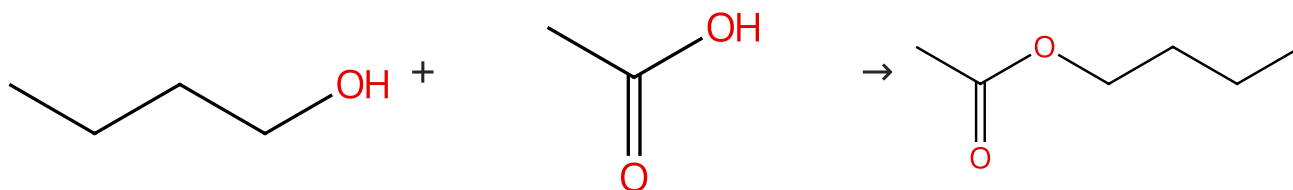
Suppliers (444)

Suppliers (91)

31-356-CAS-19595652	Steps: 1 Yield: 98%	A green and efficient solid acid catalyst from coal fly ash for Fischer esterification reaction
1.1 135 min, 95 °C		By: Mazumder, Nurul A.; et al Journal of Industrial and Engineering Chemistry (Amsterdam, Netherlands) (2015), 32, 211-217.

Scheme 34 (2 Reactions)

Steps: 1 Yield: 97-98%



Suppliers (201)

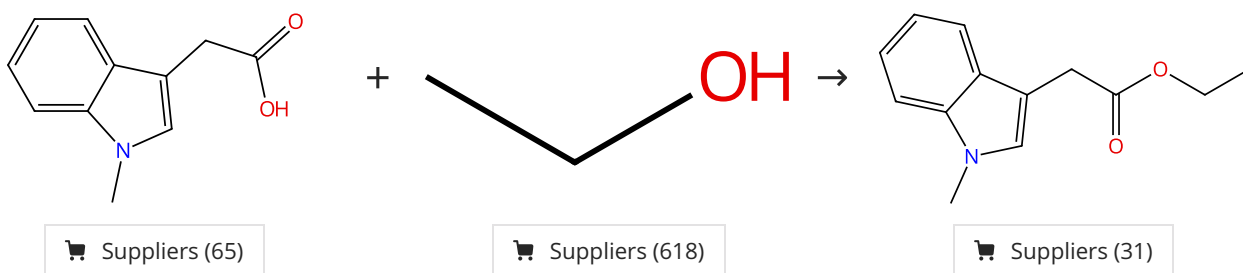
Suppliers (324)

Suppliers (142)

31-356-CAS-46600	Steps: 1 Yield: 98%	Study on the catalytic behaviors of expansible graphite in the synthesis of acetic ester and propionic ester
1.1 Catalysts: Graphite Solvents: Water; 1 h, 120 °C		By: Lin, Rui-Nian; et al Organic Chemistry: An Indian Journal (2013), 9(3), 92-96.
31-356-CAS-9769628	Steps: 1 Yield: 97%	Application of triethylammonium salts as ionic liquid catalyst and medium for fischer esterification
1.1 Catalysts: Triethylammonium sulfate Solvents: Triethylammonium sulfate; 4 h, 90 °C		By: Ganeshpure, Pralhad A.; et al ARKIVOC (Gainesville, FL, United States) (2007), (8), 273-278.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 97%



31-356-CAS-23553881

Steps: 1 Yield: 97%

Chemo- and Site-Selective Fischer Esterification Catalyzed by B(C₆F₅)₃

By: Li, Ya-Lan; et al

Asian Journal of Organic Chemistry (2021), 10(6), 1424-1427.

1.1 Catalysts: Tris(pentafluorophenyl)borane
Solvents: Ethanol; 100 h, 24 °C

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 97%



31-356-CAS-19206352

Steps: 1 Yield: 97%

Discovery and development of novel salicylate synthase (Mbt I) furanic inhibitors as antitubercular agents

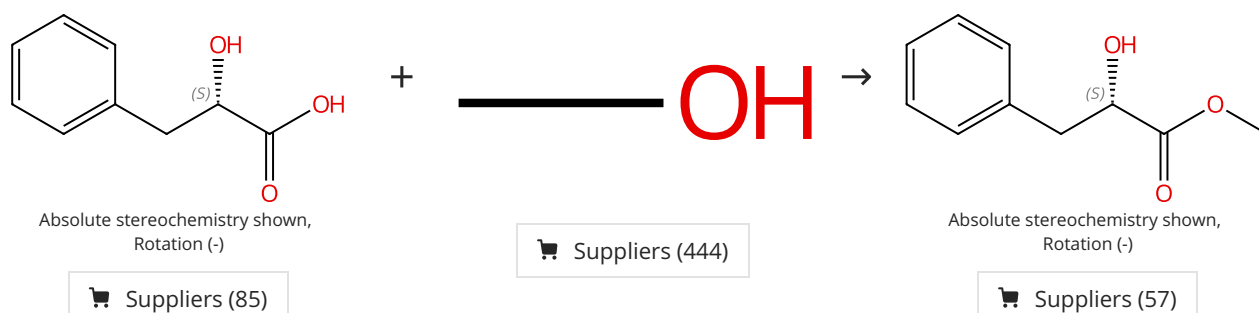
By: Chiarelli, Laurent R.; et al

European Journal of Medicinal Chemistry (2018), 155, 754-763.

1.1 Catalysts: Sulfuric acid; 24 h, reflux

Scheme 37 (1 Reaction)

Steps: 1 Yield: 97%



31-356-CAS-23553970

Steps: 1 Yield: 97%

Chemo- and Site-Selective Fischer Esterification Catalyzed by B(C₆F₅)₃

By: Li, Ya-Lan; et al

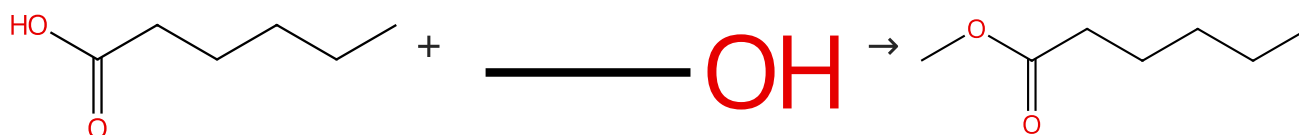
Asian Journal of Organic Chemistry (2021), 10(6), 1424-1427.

1.1 Catalysts: Tris(pentafluorophenyl)borane
Solvents: Methanol; 100 h, 24 °C

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (84)

Suppliers (444)

Suppliers (82)

31-356-CAS-11982141

Steps: 1 Yield: 97%

Solvent-free synthesis of long chain aliphatic acid methyl esters in bronsted acidic ionic liquids at room temperature

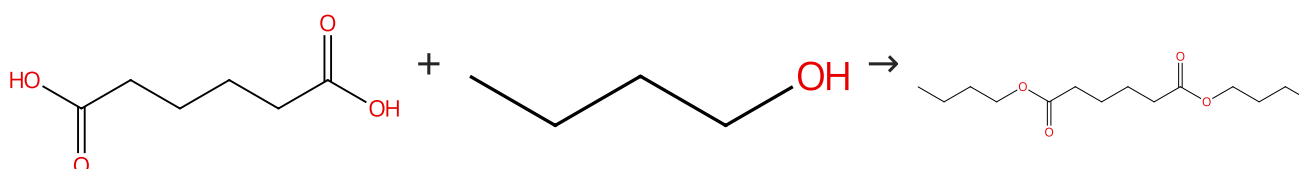
1.1 Solvents: 1*H*-Imidazolium, 1-methyl-3-[(4-sulfophenyl)methyl]-, sulfate (1:1); 3 h, rt

By: Li, Xinzhong; et al

Jingxi Huagong (2009), 26(11), 1141-1144.

Scheme 39 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (121)

Suppliers (201)

Suppliers (69)

31-356-CAS-14447508

Steps: 1 Yield: 97%

Synthesis of dicarboxylic acid esters catalyzed by silica supported sodium hydrogen sulfate

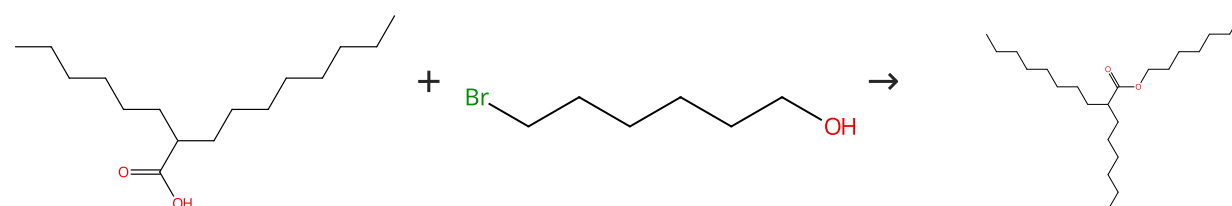
1.1 Catalysts: Sodium bisulfate
Solvents: Toluene; 90 °C

By: Xie, Xiaopeng; et al

Youji Huaxue (2010), 30(10), 1512-1516.

Scheme 40 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (57)

Suppliers (90)

Suppliers (22)

31-614-CAS-37944888

Steps: 1 Yield: 97%

Synthetic method for producing ionizable amino lipids

1.1 Catalysts: Sulfuric acid
Solvents: Cyclohexane; rt → reflux; 24 h, reflux; 24 h, reflux

Assignee: NanoVation Therapeutics Inc.

World Intellectual Property Organization, WO2023173203 A1 2023-09-21

PatentPak available

Scheme 41 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (138)

Suppliers (444)

Suppliers (102)

31-356-CAS-14102856

Steps: 1 Yield: 97%

Solvent-free synthesis of long chain aliphatic acid methyl esters in bronsted acidic ionic liquids at room temperature

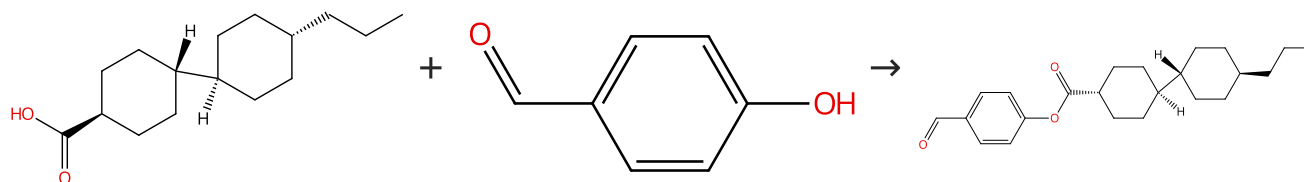
1.1 Solvents: 1*H*-Imidazolium, 1-methyl-3-[(4-sulfophenyl)methyl]-, sulfate (1:1); 3.5 h, rt

By: Li, Xinzhong; et al

Jingxi Huagong (2009), 26(11), 1141-1144.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 97%



Relative stereochemistry shown

Relative stereochemistry shown

Suppliers (52)

Suppliers (122)

31-356-CAS-1939099

Steps: 1 Yield: 97%

The Chiral Pool as Valuable Natural Source: New Chiral Mesogens Made From Lactic Acid

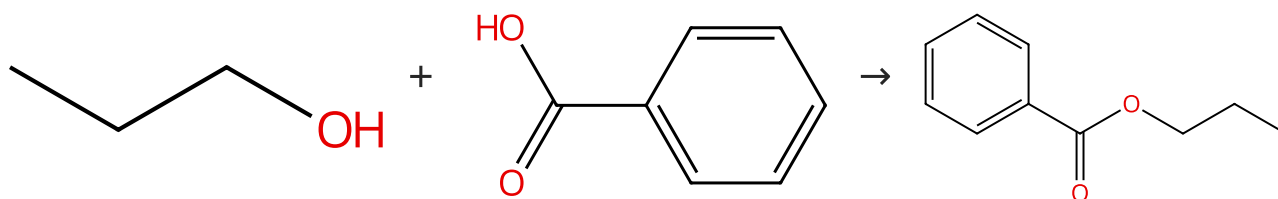
1.1 Reagents: Dicyclohexylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine

By: Brombach, F.; et al

Molecular Crystals and Liquid Crystals (2011), 542, 62-74.

Scheme 43 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (155)

Suppliers (175)

Suppliers (73)

31-614-CAS-35672321

Steps: 1 Yield: 97%

Fischer Esterification of Benzoic Acid and Polyhydric Alcohols Catalyzed by Basic Ionic Liquids of Bisimidazolium Tungstates

1.1 Catalysts: 2764700-49-2; 1 h, 0.5 M Pa, 150 °C

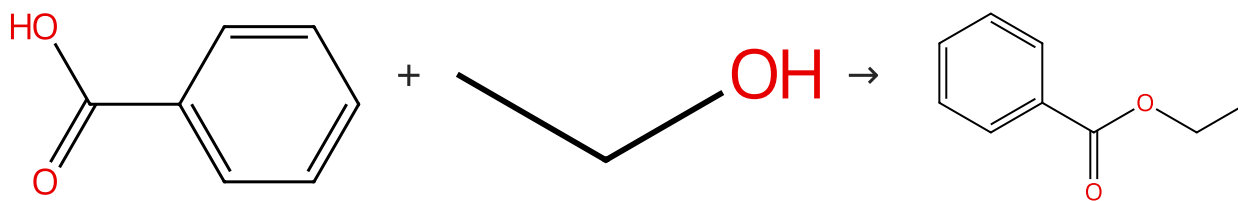
By: Wu, Shi; et al

ChemistrySelect (2023), 8(6), e202204112.

Experimental Protocols

Scheme 44 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (175)

Suppliers (618)

Suppliers (90)

31-356-CAS-14294521

Steps: 1 Yield: 97%

Novel Bronsted-acidic ionic liquids based on benzothiazolium cations as catalysts for esterification reactions

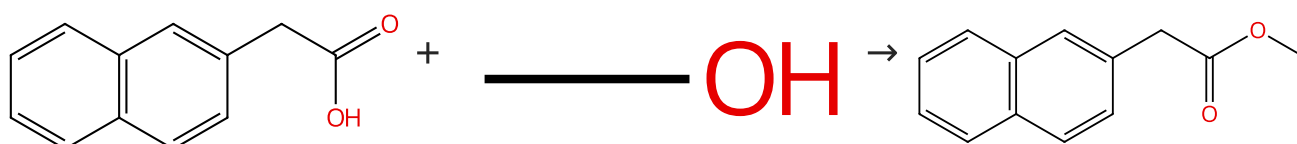
By: Zhou, Xian Si; et al

Journal of the Serbian Chemical Society (2011), 76(12), 1607-1615.

1.1 Catalysts: Benzothiazole methanesulfonate (1:1); 6 h, 110 °C; 110 °C → rt

Scheme 45 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (100)

Suppliers (444)

Suppliers (54)

31-356-CAS-23549368

Steps: 1 Yield: 97%

Chemo- and Site-Selective Fischer Esterification Catalyzed by B(C₆F₅)₃

By: Li, Ya-Lan; et al

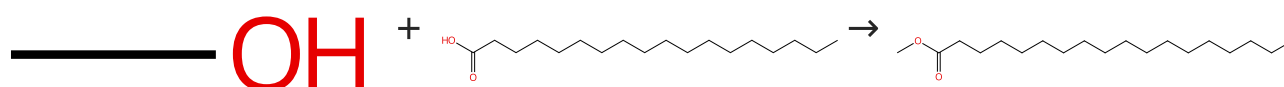
Asian Journal of Organic Chemistry (2021), 10(6), 1424-1427.

 1.1 Catalysts: Tris(pentafluorophenyl)borane
 Solvents: Methanol; 100 h, 24 °C

Experimental Protocols

Scheme 46 (2 Reactions)

Steps: 1 Yield: 97%



Suppliers (444)

Suppliers (155)

Suppliers (108)

31-356-CAS-17410224

Steps: 1 Yield: 97%

Efficient flow Fischer esterification of carboxylic acids with alcohols using sulfonic acid-functionalized silica as supported catalyst

By: Furuta, Akihiro; et al

Bulletin of the Chemical Society of Japan (2017), 90(5), 607-612.

 1.1 Catalysts: 2-Hydroxy-3-[3-(trihydroxysilyl)propoxy]-1-propanesulfonic acid (silica gel supported)
 Solvents: Methanol, Tetrahydrofuran; 23 min, 75 psi, 110 °C

Experimental Protocols

31-356-CAS-5588375	Steps: 1 Yield: 97%	Novel Bronsted acidic deep eutectic solvent as reaction media for esterification of carboxylic acid with alcohols
1.1 Catalysts: <i>p</i> -Toluenesulfonic acid, 1-Octanaminium, <i>N,N,N</i> -trimethyl-, methanesulfonate (1:1); 10 min, rt → 60 °C		By: De Santi, Valerio; et al
1.2 2 h, rt → 60 °C		Tetrahedron Letters (2012), 53(38), 5151-5155.
Experimental Protocols		

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